HACPO. Optimized Specifcation

This fle refnes the original HACPO (Hybrid Adaptive Contrastive Policy Optimization) objective. we 1) unify notation, 2) make clipping/PPO mechanics explicit, 3) defne the contrastive term and weights, 4) give practical hyperparameter suggestions and 5) provide implementation notes and pseudocode.

Notation (unifed).

- θ. policy parameters for $\pi_\theta(y \mid x)$.
- π_{ref} . reference / behavior policy (fxed when computing ratios).
- $r_i(\theta) = \dfrac{\pi_\theta(y_i \mid x_i)}{\pi_{\text{ref}}(y_i \mid x_i)}$. importance / policy ratio for supervised sample (x_i, y_i).
- $r_j^{(k)}(\theta) = \dfrac{\pi_\theta(y_{j,k} \mid x_j)}{\pi_{\text{ref}}(y_{j,k} \mid x_j)}$. ratio for the k-th contrastive (negative) sample associated with context x_j.
- $\hat{A}_i$. estimated (possibly generalized) advantage for sample i. Use the same notation for all advantages for clarity.
- $w_{j,k} \geq 0$. weight for contrastive sample $y_{j,k}$; we require $\sum_k w_{j,k} = 1$ for each context j.
- Clip interval parameter $\epsilon > 0$; defne $\text{clip}_\epsilon(r) = \text{clip}(r, 1-\epsilon, 1+\epsilon)$.
- L_{CPO}. contrastive policy optimization loss (defned below).
- $D_{\text{KL}}(\pi_{\text{ref}} \| \pi_\theta)$. KL divergence used as a constraint/penalty. (See note about direction and sign.)

Objective

We present the HACPO objective in a form consistent with PPO-style clipping and with an explicit contrastive term.

$$
\begin{aligned}
J_{\text{HACPO}}(\theta) = \mathbb{E}_{(x_i,y_i)\sim\mathcal{D}} \Bigg[& \alpha_i \cdot \underbrace{\min\Big(r_i(\theta)\,\hat{A}_i,\ \text{clip}_\epsilon(r_i(\theta))\,\hat{A}_i\Big)}_{\text{supervised/PPO-clip term}} \\
& + (1-\alpha_i)\cdot \underbrace{\sum_{k=1}^{K} w_{i,k} \min\Big(r_i^{(k)}(\theta)\,\hat{A}_i^{(k)},\ \text{clip}_\epsilon(r_i^{(k)}(\theta))\,\hat{A}_i^{(k)}\Big)}_{\text{contrastive importance-sampled PPO-clip}} \Bigg] \qquad (1) \\
& + \gamma\,\mathbb{E}_{x_i\sim\mathcal{D},\,\{y_{i,k}\}\sim A}\big[\,L_{\text{CPO}}(\theta;\, x_i, \{y_{i,k}\})\big] \\
& - \beta_{\text{KL}} D_{\text{KL}}\big(\pi_{\text{ref}} \,\|\, \pi_\theta\big).
\end{aligned}
$$

Notes on signs and KL direction. Many RL formulations either maximize an objective (PPO) or minimize a loss. Above we write J_{HACPO} as an objective to maximize; the KL term is written as a penalty by subtracting $\beta_{\text{KL}} D_{\text{KL}}(\pi_{\text{ref}} \| \pi_\theta)$. If you implement using gradient descent on a loss, negate J_{HACPO} accordingly. We use $D_{\text{KL}}(\pi_{\text{ref}} \| \pi_\theta)$ so that the penalty encourages π_θ to stay close to the reference policy [?].

Contrastive loss. L_{CPO}

One practical choice for the contrastive policy loss (inspired by InfoNCE-style contrastive objectives, adapted to policies) is.

$$
L_{\text{CPO}}(\theta; x, \{y_k\}) = -\log \frac{\exp\big(s_\theta(x, y^+)/\tau_c\big)}{} \qquad (2)
$$